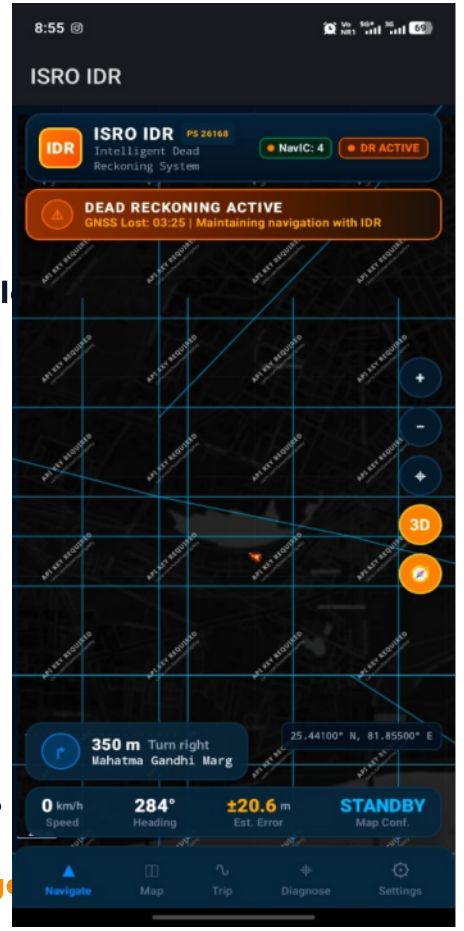
SMART INDIA HACKATHON 2026

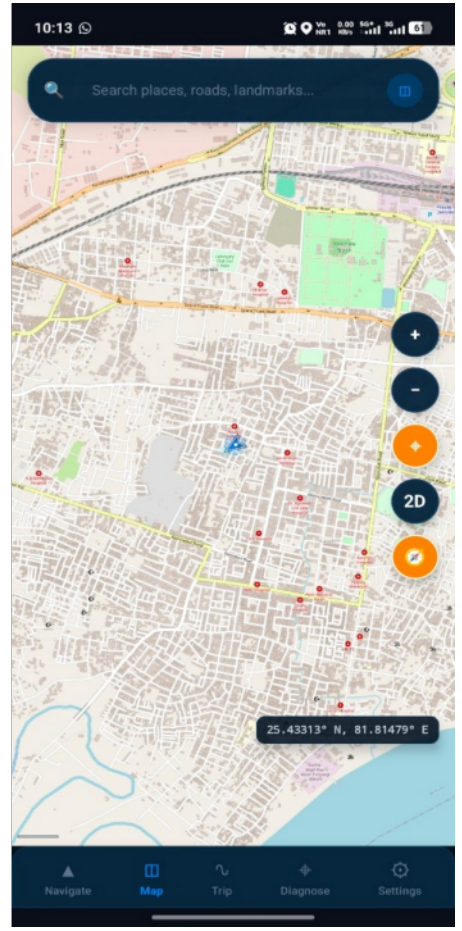
TITLE PAGE — OFFICIAL SUBMISSION

Problem Statement ID: 26168

Problem Statement Title:	**AI-ML based Intelligent Dead Reckoning system for seaml**
Theme:	**Smart Vehicles**
PS Category:	**Software**
Organization / Ministry:	Indian Space Research Organisation (ISRO) / Department of Space
Team ID:	**[Registered Team ID on Portal]**
Team Name:	**[Registered Team Name on Portal]**
Idea Title:	**ISRO IDR — Intelligent Dead Reckoning for Seamless**
Product Tagline:	**AI-Powered Resilient Inertial Navigation During GNSS Outage**



Main Navigation HUD



Live NavIC SkyView

IDEA TITLE: ISRO IDR — RESILIENT NAVIGATION DURING GNSS OUTAGES

• Proposed Solution (Idea/Prototype) • Detailed Explanation • How it Addresses Problem • Innovation & Uniqueness

1. THE PROBLEM: GNSS BLACKOUT

- Signal Blockage: Tunnels, underpasses & urban canyons cause complete GNSS loss-of-lock.
- Multipath Jumps: High-rise reflections create false position jumps (>50 m) at tunnel exits.
- 3D Layer Failure: 2D maps erroneously snap flyover vehicles onto ground roads (0% layer accuracy).
- Raw IMU Drift: Consumer phone MEMS drifts O(t^3), exceeding 75% error in <30s without wheel sensors.

2. PROPOSED SOLUTION & PIPELINE

- Auto In-Vehicle Alignment: Gravity leveling + PCA forward heading calibrates arbitrary phone angles.
- Vibration Pre-Filter: 45 spectral moments blank engine idling and pothole suspension shocks.
- Transpiled GBDT Speed: Predicts forward velocity from IMU without OBD-II/CAN bus (R^2 = 0.884).
- 15-State ESKF: Quaternion attitude + NHC (vy, vz ≈ 0) and standstill ZUPT clamping.
- Causal 3D HMM: |Δh| > 4m elevation barrier prevents impossible flyover layer transitions.
- Deficit Supervisor: Chi-Square innovation gate (γ ≤ 11.35) manages seamless GNSS <-> DR transition.

3. INNOVATION & UNIQUENESS

- Zero External ML Runtimes: GBDT transpiled to static C++ decision tables (No PyTorch/TFLite/ONNX).
- 99.7%–100% Flyover Accuracy: 3D HMM solves the multi-level road snapping failure.
- Dual-Profile Engine: 10 Hz Android Mobile App + >50 kHz external IMU C++ CLI.
- Native NavIC Ingestion: Ingests Android GnssStatus for live IRNSS satellite radar.
- 100% Offline Capability: Compact binary .idrmap (1.2 MB/corridor) requires zero API keys.
- Traceable Validation: Python research (27.25% med) + Frozen N7 C++ production reference (35.71%).

TECHNICAL APPROACH: FROM RAW SENSORS TO RESILIENT POSITION

• Technologies (C++17, Python, Kotlin, Compose, JNI) • Methodology Flow • Working Mobile Prototype

SYSTEM ARCHITECTURE & CAUSAL PROCESSING FLOW

1. Multi-Sensor Ingestion Layer (~100 Hz IMU / ~200 Hz External IMU + 1 Hz GNSS/NavIC):
   Ingests raw accelerometer, gyroscope, magnetometer, barometer, and GnssStatus callbacks.

2. Preprocessing & Auto-Alignment Engine:
   Gravity leveling calculates pitch/roll; PCA forward acceleration resolves vehicle yaw mounting matrix R_p^b.

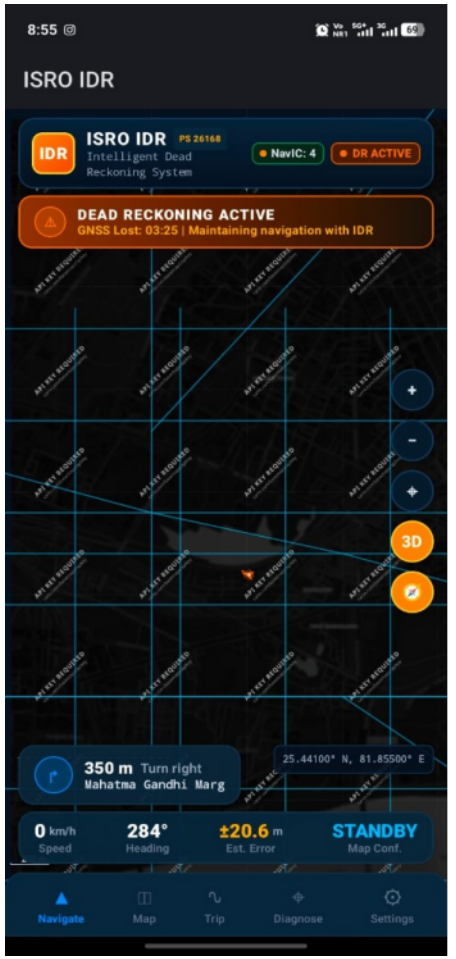
3. AI Velocity & Vibration Filter (GBDT Regressor):
   120 branchless trees on 1.0s window (45 spectral moments) predict forward speed ($R^2 = 0.884$, RMSE 1.21 m/s).

4. 15-State Continuous-Discrete ESKF Fusion:
   State $x = [p, v, q, b_a, b_g]$ with Non-Holonomic Constraints ($v_y, v_z \approx 0$) and standstill ZUPT clamping.

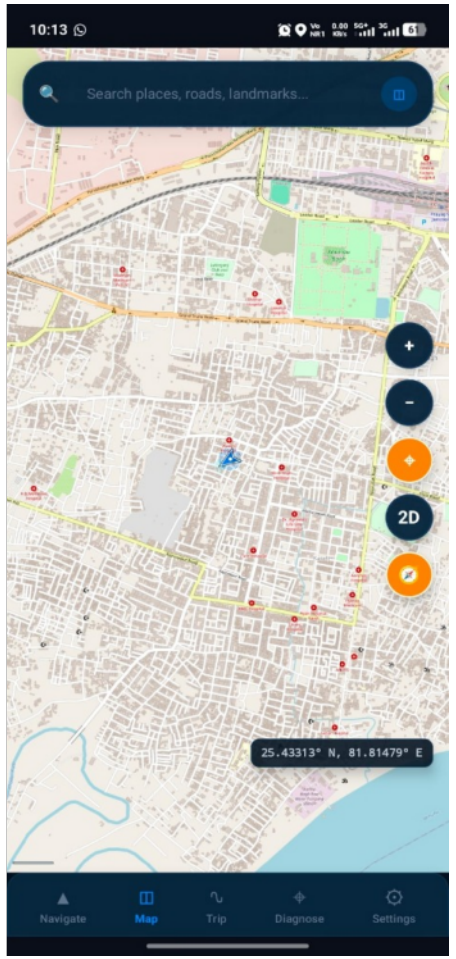
5. Causal 3D HMM Map Matcher ($|\Delta h| > 4.0$m Elevation Barrier):
   Viterbi decoding projects position onto topological .idrmap graph, guaranteeing 99.7%-100% flyover layer accuracy.

6. Deficit Supervisor & Delivery:
   Chi-Square gate ($\gamma \leq 11.35$) rejects multipath -> 10 Hz Android UI StateFlow / >50 kHz Edge C++ CLI.

TECH STACK: C++17 Core | Python 3.11 | Kotlin & Compose | JNI | Valhalla/OSM Vector Engine | Android GnssStatus



Main Navigation HUD



Live NavIC Radar

FEASIBILITY & VIABILITY: ENGINEERED FOR REAL-TIME & EDGE OPERATION

• Quantitative Feasibility Metrics • Key Challenges & Risks • Engineering Strategies to Overcome

1. MEASURED FEASIBILITY METRICS

• 56,026 Hz Edge Throughput:
>280x required 200 Hz tactical rate.

• 17.85 µs Filter Latency:
Sub-millisecond per-sample step.

• <2.4 MB RSS Memory:
Zero dynamic heap growth in stream.

• 99.7%–100% Flyover Accuracy:
Eliminates ground road snapping.

• R^2 = 0.884 AI Speed Regressor:
RMSE 1.21 m/s on real vehicle logs.

• 5,101 Bit-for-Bit Parity Vectors:
11.8 picometer max C++ delta.

2. KEY CHALLENGES & RISKS

• Sensor Bias & Inertial Drift:
Phone MEMS IMUs drift with O(t^3)
without wheel tick feedback.

• Chassis & Engine Vibration:
Road shocks corrupt acceleration.

• Multi-Level Road Ambiguity:
Parallel elevated roads confuse 2D.

• Mobile Resource Limits:
Heavy ML models cause thermal throttle.

3. ENGINEERING STRATEGIES

• AI Velocity + NHC/ZUPT:
Constrains velocity & clamps stationary
drift, mitigating bias.

• 45-Feature Pre-Filter:
Isolates motion from vibration harmonics.

• 3D HMM Elevation Barrier:
|Δh| > 4.0m threshold blocks hops.

• Transpiled C++17 Core:
Branchless tables run with zero ML runtime.

T BENCHMARK BOUNDARY: Short/Medium Outages achieve 1.62%–2.88% drift (Exceeding target on normal corridors). Full 158-outage dataset-wide <10% drift remains an open research cha

IMPACT & BENEFITS: RESILIENT POSITIONING BEYOND SATELLITES

• Potential Impact on Target Audiences • Social, Economic & Strategic Benefits

AUTOMOTIVE & FLEET LOGISTICS

Eliminates navigation freezing in tunnels, underpasses, and urban high-rise corridors. Prevents signal loss-induced route deviations and enables seamless last-mile delivery continuity.

SMART MOBILITY & TWO-WHEELERS

Enhances rider safety and navigation reliability in dense urban canyons, flyovers, and underpasses where satellite signals degrade or disappear.

DEFENCE, AEROSPACE & ROBOTICS

Edge-deployable standalone C++ engine for tactical UGVs, autonomous drones, and military vehicles, ensuring mission-critical navigation in GNSS-denied environments.

SOVEREIGN NAVIC PROMOTION

Directly integrates and processes NavIC / IRNSS constellation signals on mobile devices, promoting indigenous satellite adoption and reducing dependence on foreign GNSS systems.

CORE VALUE: RESILIENT CONTINUITY | ZERO HARDWARE RETROFIT | 100% OFFLINE | NAVIC-READY | EDGE & MOBILE DUAL PROFILE

RESEARCH & REFERENCES: FROM ALGORITHM TO VERIFIED MVP

• Reference & Research Work • 158-Outage IO-VNBD Benchmark Progression • Authoritative Evidence

RESEARCH PROGRESSION (158 REAL OUTAGES)

- C0 Baseline (Pure Strapdown DR): 76.34% Median (12/158 pass)
- N0 Reference (AI Speed + Basic ESKF): 34.43% Median (28/158 pass)
- N7 Frozen Production C++: 35.71% Median (36/158 pass)
- Phase 14 Feature Pruning: 31.65% Median (39/158 pass)
- Phase 15 Jerk Derivatives: 30.88% Median (39/158 pass)
- Phase 16 Doppler Handover: 27.25% Median (45/158 pass) — Best Median
- Phase 17 Kinematic Consistency: 27.64% Median (49/158 pass) — Best Passes

AUTHORITATIVE EVIDENCE & REFERENCES

- Mandated IO-VNBD Dataset:
  Onyekpe et al., 'IO-VNBD: Inertial & Odometry Benchmark Dataset for Ground Vehicles' (Evaluated on all 158 authentic outages).
- 15-State ESKF & AI-IMU Formulation:
  J. Solà, 'Quaternion kinematics for error-state KF' (2017) & M. Brossard et al., 'AI-IMU Dead-Reckoning' (IEEE T-RO 2020).
- Real-World Field Validation:
  3.2 km Prayagraj corridor trial with u-blox ZED-F9P RTK-GNSS.
- Edge Engine Performance:
  56,026 Hz streaming throughput, 17.85 µs latency, <2.4 MB RSS.
- Repository Evidence Base:
  266 unit/integration tests passing; 14 canonical forensic reports in docs/canonical/ at frozen commit a17a88b.

RCH INTEGRITY: Phase 16/17 models belong to Python research (27.25% med / 49 passes); C++ core executes the frozen N7 baseline (35.71% med / 36 passes). Realistic corridors achieve 1.62%–2.88%